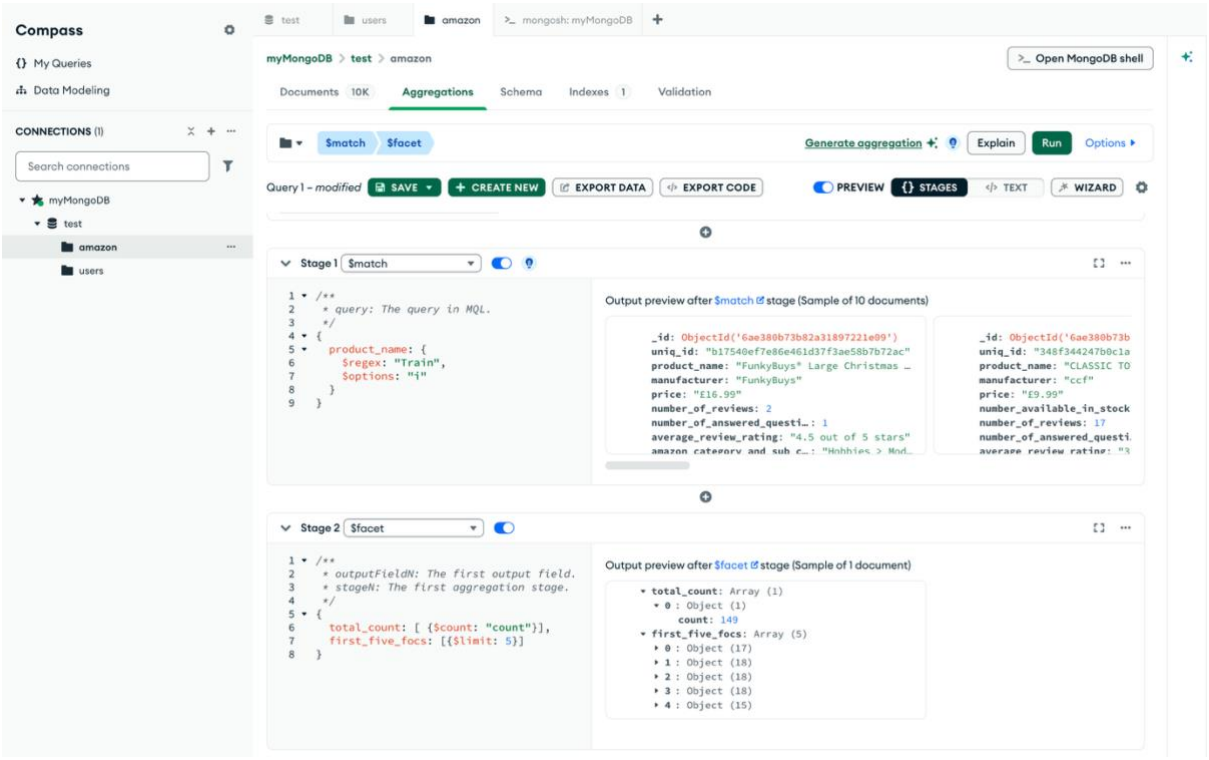
Dr. Forouraghi

CSC 351 D01

Dan Shapovalov

Homework #6

Query 1 MongoDB Compass



Query 1 mongosh

```
db.amazon.aggregate([
  { $match: {
      product_name: { $regex: "Train", $options: "i" }
  }},
  { $facet: {
      total_count: [{ $count: "count" }],
      first_five_docs: [{ $limit: 5 }]
  }}
]);
```

Query 1 connection to SQL

db.amazon = FROM amazon

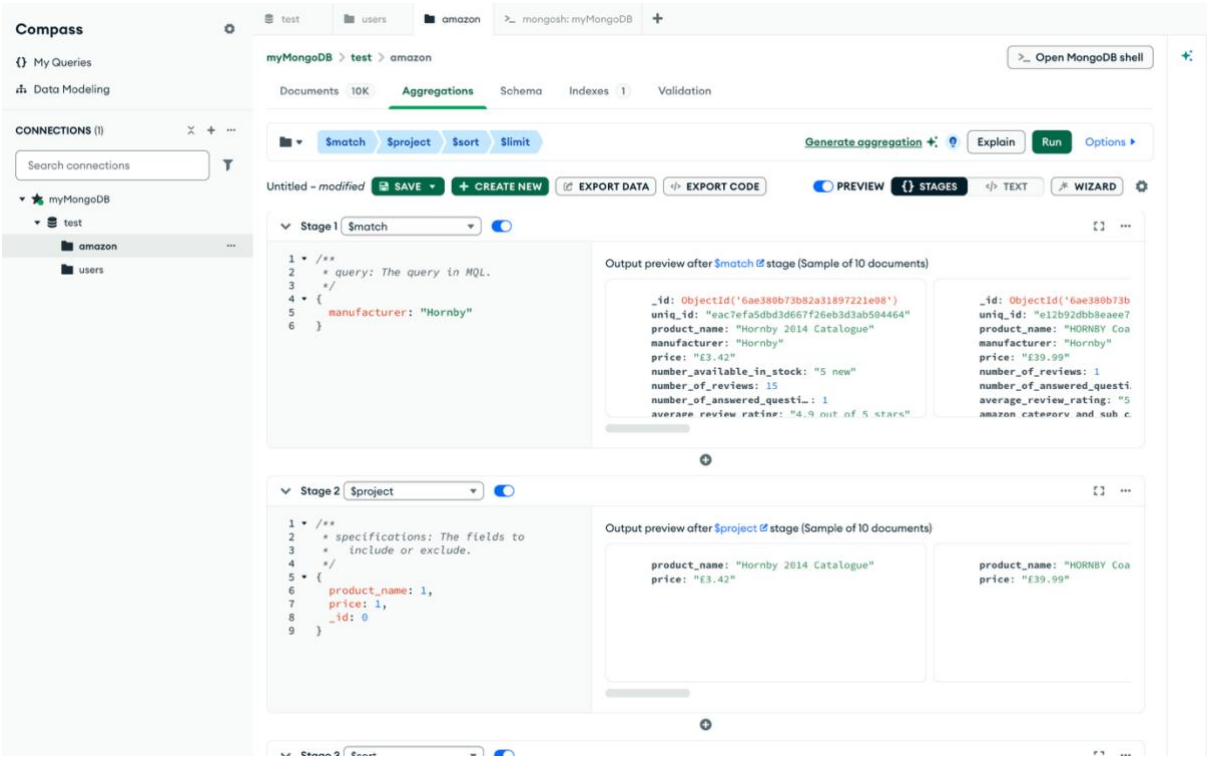
{ $match: {

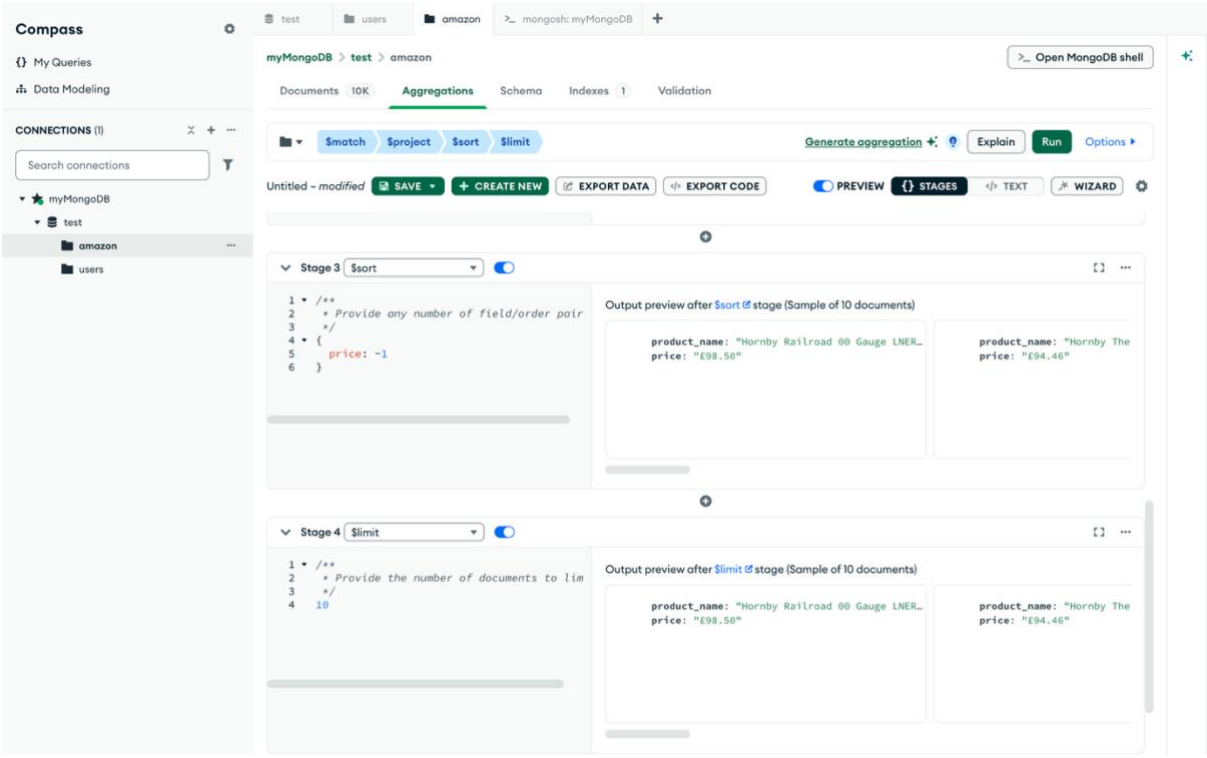
product_name: { $regex: "Train", $options: "i" }

}}, = WHERE product_name LIKE '%Train%'

first_five_docs: [{ $limit: 5 }] = LIMIT 5

Query 2 MongoDB Compass





Query 2 mongosh

db.amazon.find({manufacturer: "Hornby"}, {_id: 0, product_name: 1, price: 1}).sort({price: -1}).limit(10)

Query 2 connection to SQL

db.amazon = FROM amazon

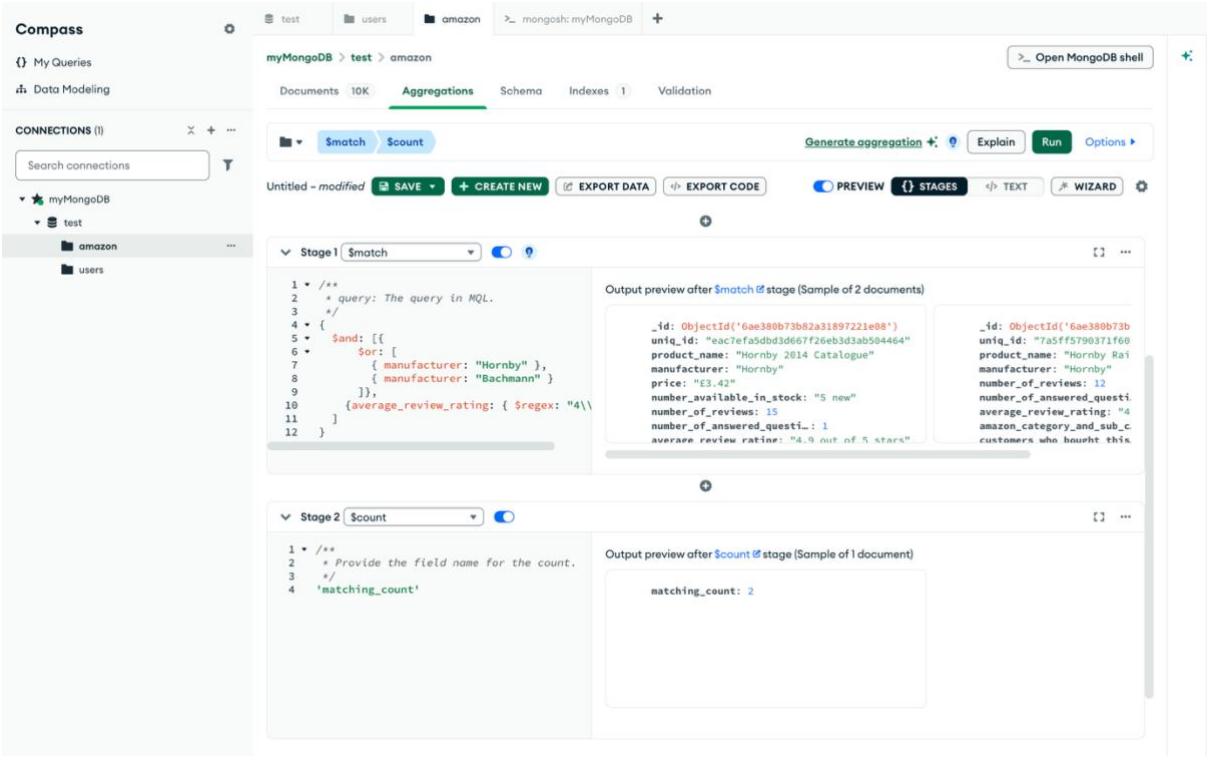
{_id: 0, product_name: 1, price: 1} = SELECT product_name, price

find({manufacturer: "Hornby"} = WHERE manufacturer = ‘Hornby’

sort({price: -1}) = ORDER BY price DESC

limit(10) = LIMIT 10

Query 3 MongoDB Compass



Query 3 mongosh

```
db.amazon.aggregate([
 { $match: {
    $or: [
     { manufacturer: "Hornby" },
     { manufacturer: "Bachmann" }
    ],
    average_review_rating: { $regex: "4\\.9" }
 }},
 { $count: "matching_count" }
]);
```

Query 3 connection to SQL

db.amazon = FROM amazon

{ $match: {

$or: [

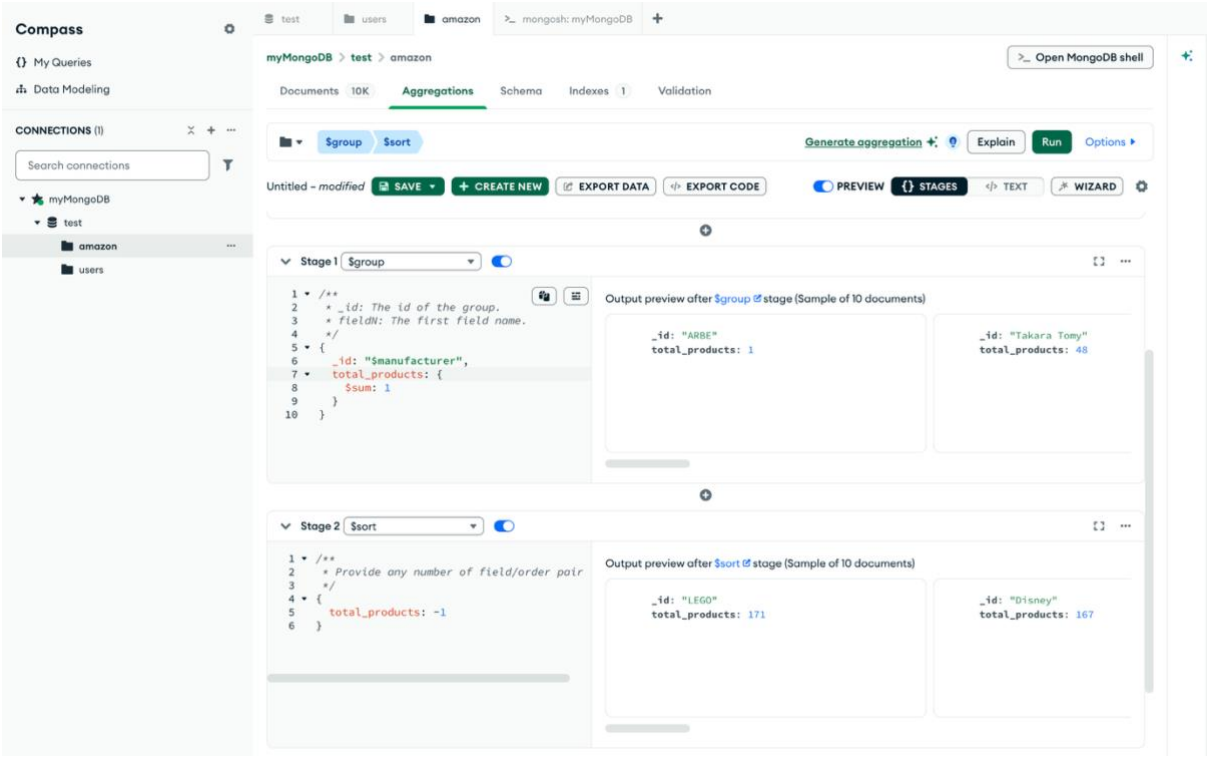
{ manufacturer: "Hornby" },

{ manufacturer: "Bachmann" }

] = WHERE (manufacturer = ‘Hornby’ OR manufacturer = ‘Bachmann’)

average_review_rating: { $regex: "4\\.9" } = average_review_rating LIKE '%4.9%'

Query 4 MongoDB Compass



Query 4 mongosh

```
db.amazon.aggregate([
 { $group: {
    _id: "$manufacturer",
    total_products: { $sum: 1 }
 }},
 { $sort: { total_products: -1 } }
]);
```

Query 4 connection to SQL

```
SELECT manufacturer, COUNT(*) AS total_products
FROM amazon
GROUP BY manufacturer
ORDER BY total_products DESC;
```

db.amazon = FROM amazon

total_products: { $sum: 1 } = COUNT(*) AS total_products

$group: {

_id: "$manufacturer", = GROUP BY manufacturer

{ $sort: { total_products: -1 } } = ORDER BY total_products DESC